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Université: Institut Universitaire des Sciences (IUS)

TD N° 5: Réseau1

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Niveau: 3^{ème} Année

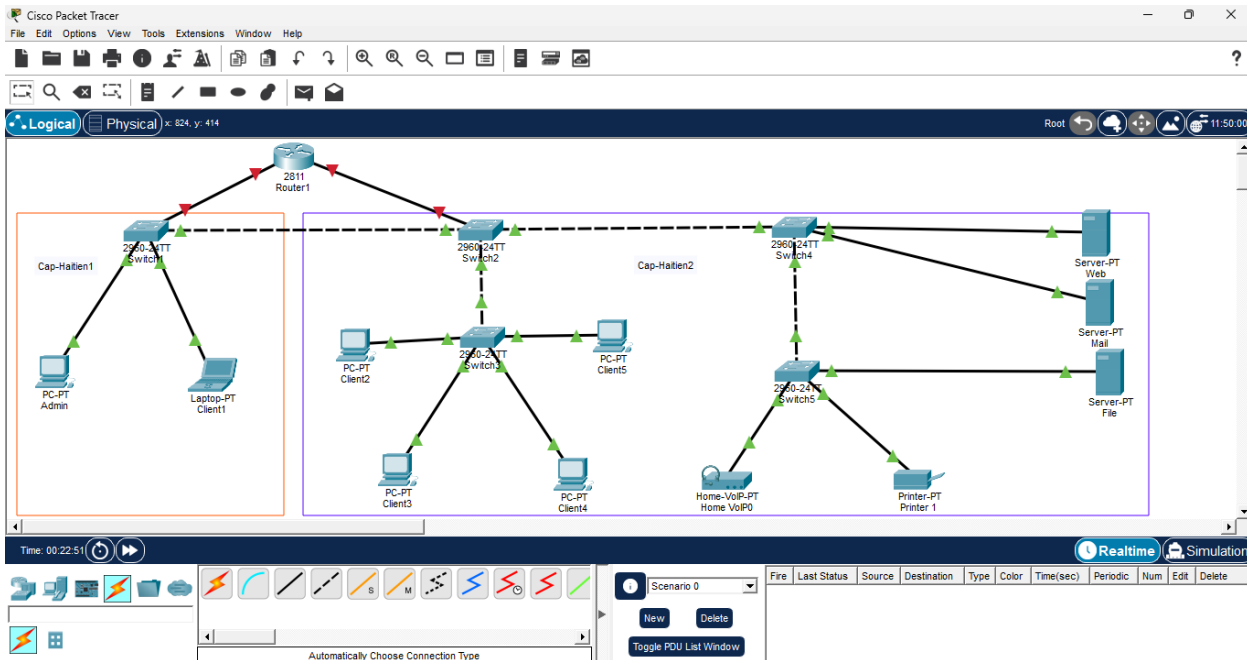
Date: L 08 /Janvier/2025

Objectif du TD

L'objectif de ce TD est de reproduire une topologie réseau fonctionnelle, puis de configurer les services DNS, HTTP et HTTPS afin de permettre l'accès sécurisé et non sécurisé à une page web. Ce travail permet de comprendre le rôle de chaque service réseau, d'assurer la communication entre les différents équipements, et de vérifier le bon fonctionnement du réseau à travers des tests de connectivité et d'accès web.

1. Reproduisez cette topologie en configurant les services DNS, HTTP et HTTPS, puis en affichant une page web.

a) J'ai reproduit cette topologie en configurant les switches



- Sur cette image, j'ai reproduit la topologie demandée.

1.1 J'ai attribué les adresses IP aux dispositifs en Utilisant IPv4

a) Un exemple de IP PC Client2: 192.168.2.6

The screenshot shows the 'Client2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the 'FastEthernet0' interface. The 'Static' radio button is selected for IPv4 configuration. The IPv4 Address is set to 192.168.2.6, and the Subnet Mask is set to 255.255.255.0. The Default Gateway and DNS Server are both set to 0.0.0.0. The IPv6 Configuration section is also visible, with the 'Static' radio button selected. The IPv6 Address is empty, and the Link Local Address is set to FE80::290:2BFF:FE0C:6AA0. The 802.1X section is also visible, with the 'Use 802.1X Security' checkbox unchecked. The Authentication dropdown is set to MD5, and the Username and Password fields are empty. A 'Top' button is located at the bottom left of the window.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.2.6
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::290:2BFF:FE0C:6AA0
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

☐ Top

- Sur cette image, j'ai configuré l'adresse IP et le masque de sous-réseau (Mask Subnet) du **Client2**.

-

b) Un exemple de IP pour Client4

Client4: 192.168.2.8

The screenshot shows a window titled "Client4" with a tabbed interface. The "Desktop" tab is selected, displaying the "IP Configuration" section. The "Interface" dropdown is set to "FastEthernet0". Under "IP Configuration", the "Static" radio button is selected. The fields are filled with: IPv4 Address: 192.168.2.8, Subnet Mask: 255.255.255.0, Default Gateway: 0.0.0.0, and DNS Server: 0.0.0.0. The "IPv6 Configuration" section has "Static" selected, with IPv6 Address, Link Local Address (FE80::260:47FF:FE80:4932), Default Gateway, and DNS Server fields. The "802.1X" section has "Use 802.1X Security" unchecked, with Authentication set to "MD5" and empty Username and Password fields. A "Top" button is at the bottom left.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.2.8
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::260:47FF:FE80:4932
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

☐ Top

- Sur cette image, j'ai configuré l'adresse IP et le masque de sous-réseau (Mask Subnet) du **Client4**.

c) Un exemple de IP pour le Server Web

Serveur Web: 192.168.2.10

The screenshot shows a configuration window titled 'Web' with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Desktop' tab is selected, and the 'IP Configuration' section is active. It contains three main configuration areas: IP Configuration, IPv6 Configuration, and 802.1X.

IP Configuration

- ☐ DHCP
- ☒ Static
- IPv4 Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 0.0.0.0
- DNS Server: 0.0.0.0

IPv6 Configuration

- ☐ Automatic
- ☒ Static
- IPv6 Address: [Empty field] / [Empty field]
- Link Local Address: FE80::2D0:BCFF:FECB:3AB6
- Default Gateway: [Empty field]
- DNS Server: [Empty field]

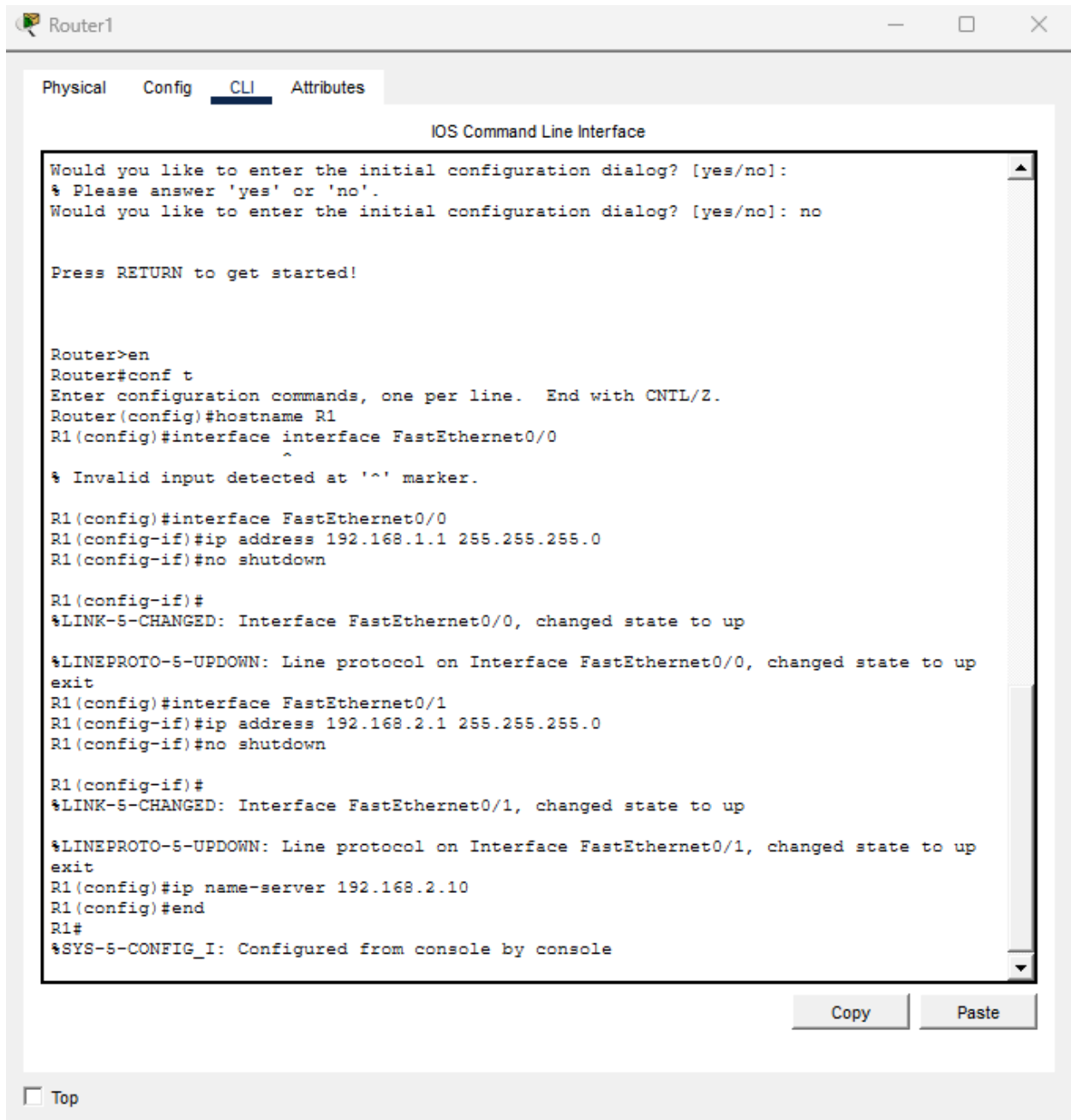
802.1X

- ☐ Use 802.1X Security
- Authentication: MD5 (dropdown menu)
- Username: [Empty field]
- Password: [Empty field]

At the bottom left, there is a 'Top' button.

- Sur cette image, j'ai configuré l'adresse IP et le masque de sous-réseau du serveur Web.

a) J'ai configuré le routeur



- Sur cette image, j'ai configuré le routeur1.

- Pour les switches, j'ai sélectionné un échantillon de trois switches (1, 2 et 3).

The screenshot shows a window titled "Switch1" with tabs for Physical, Config, CLI (selected), and Attributes. The CLI tab displays the "IOS Command Line Interface". The text in the terminal window is as follows:

```

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#interface vlan 1
SW1(config-if)#ip address 192.168.1.2 255.255.255.0
SW1(config-if)#no shutdown

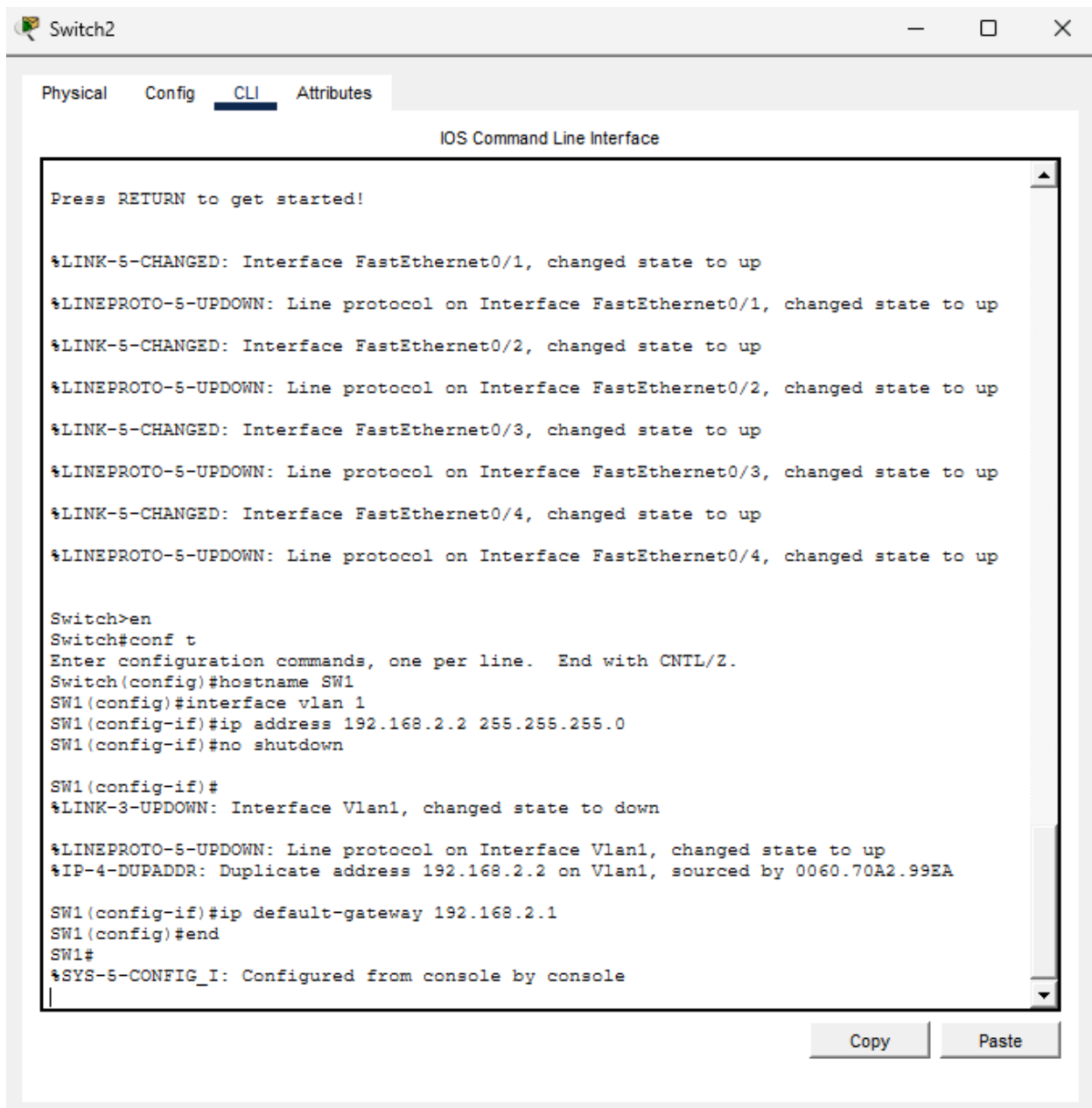
SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

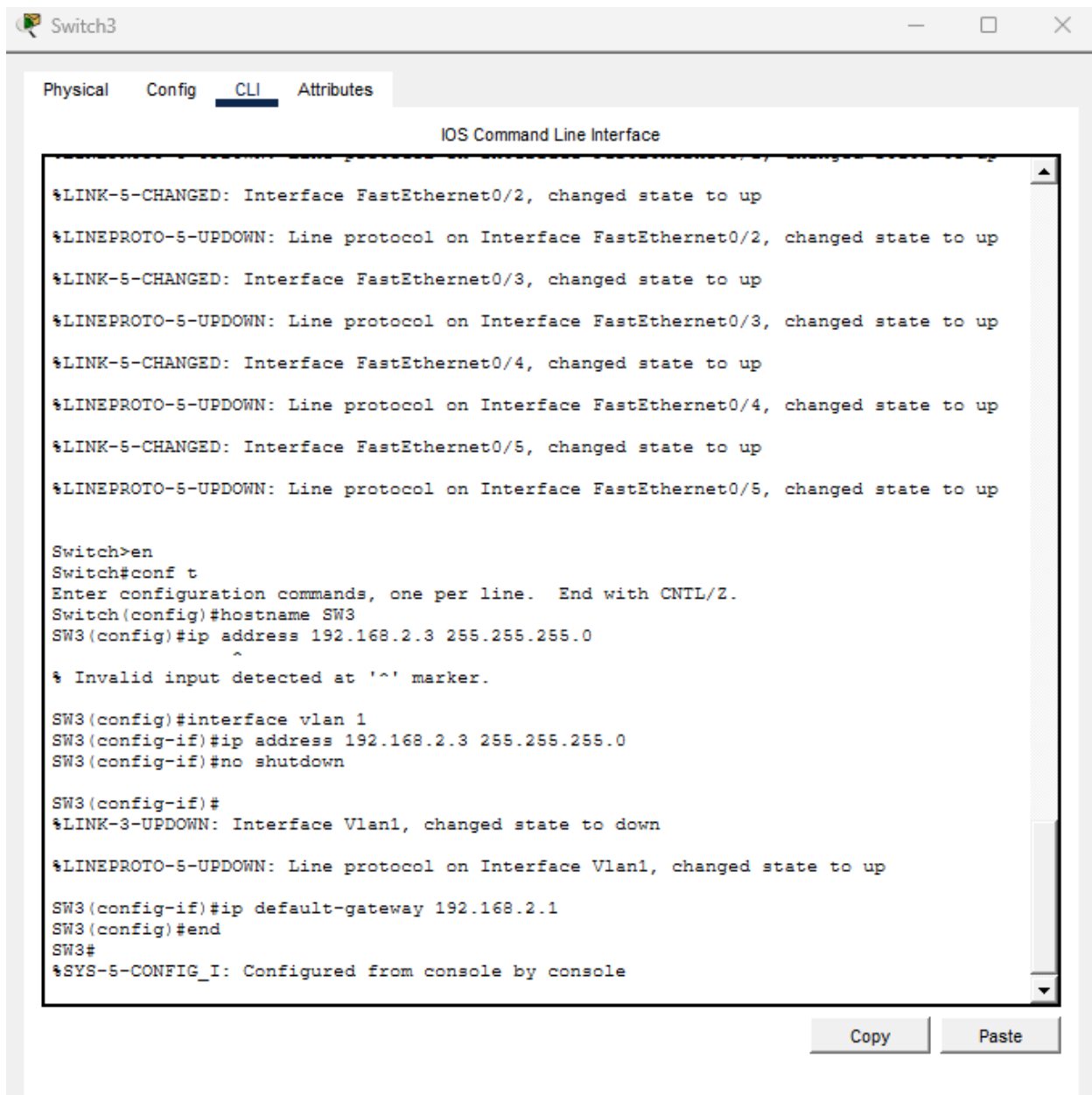
SW1(config-if)#ip default-gateway 192.168.1.1
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
|
  
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons.

- Sur cette image, j'ai configuré le Switch1 server du Server File.

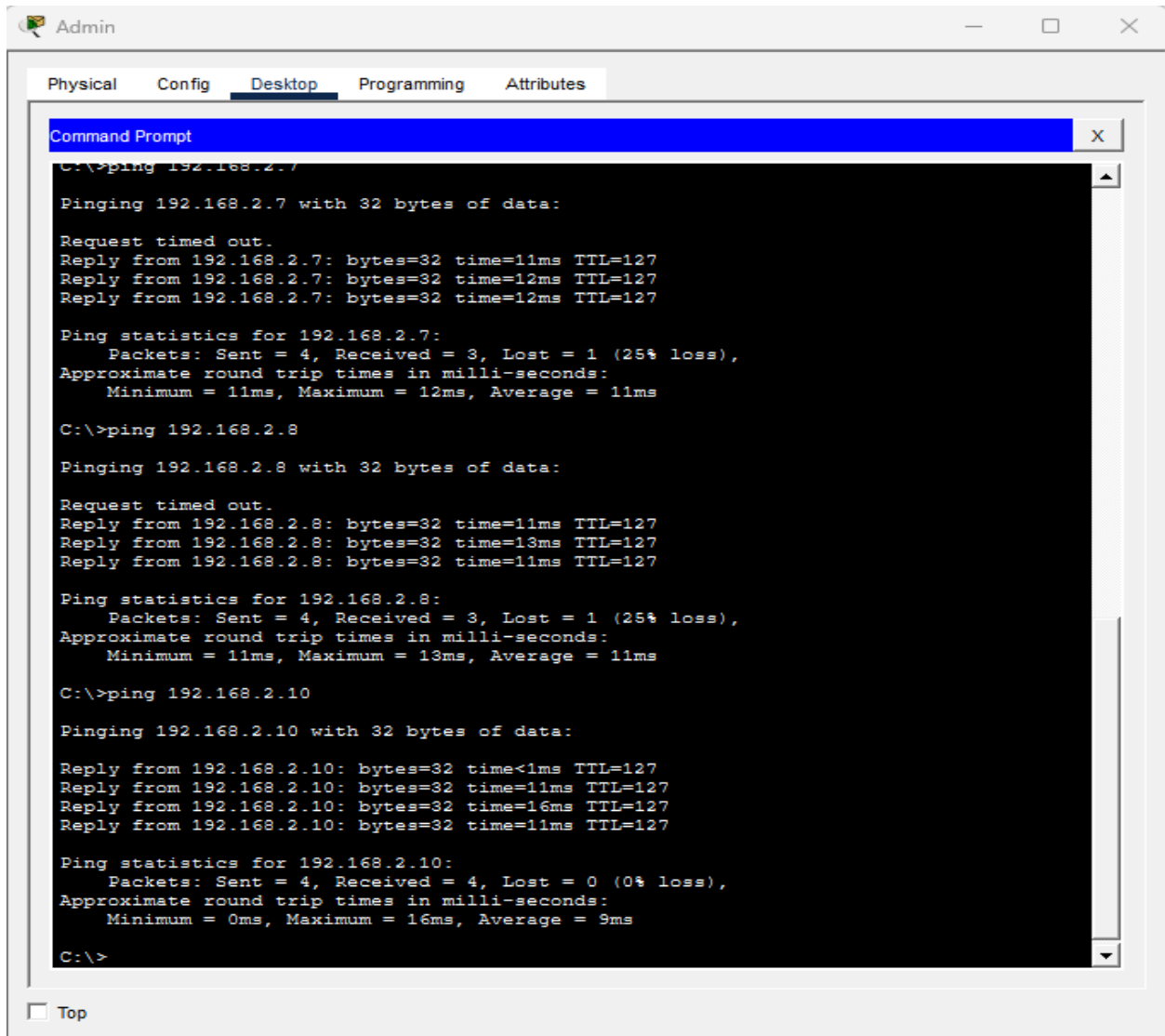


- Sur cette image, j'ai configuré le Switch2 .



- Sur cette image, j'ai configuré le Switch3.

b) J'ai testé la connectivité avec la commande ping (Tester la connectivité entre les hôtes).



The screenshot shows a window titled "Admin" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of three ping commands executed from the C:\> directory.

```
C:\>ping 192.168.2.7

Pinging 192.168.2.7 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.7: bytes=32 time=11ms TTL=127
Reply from 192.168.2.7: bytes=32 time=12ms TTL=127
Reply from 192.168.2.7: bytes=32 time=12ms TTL=127

Ping statistics for 192.168.2.7:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 12ms, Average = 11ms

C:\>ping 192.168.2.8

Pinging 192.168.2.8 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.8: bytes=32 time=11ms TTL=127
Reply from 192.168.2.8: bytes=32 time=13ms TTL=127
Reply from 192.168.2.8: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.2.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 13ms, Average = 11ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

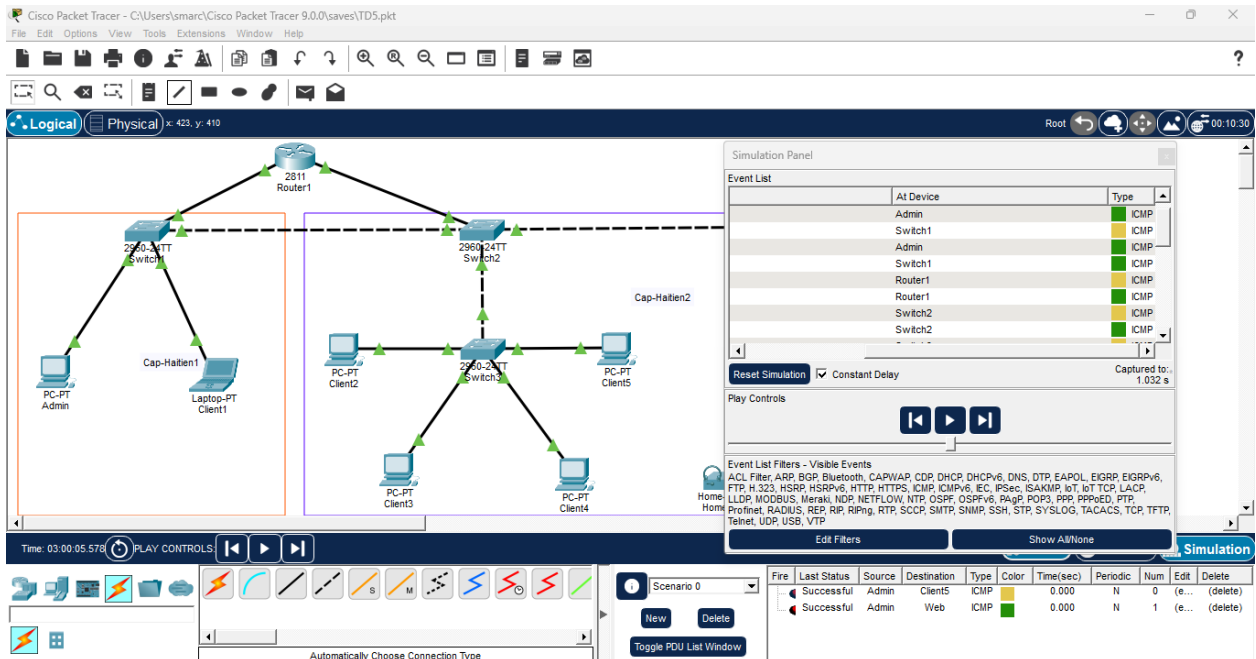
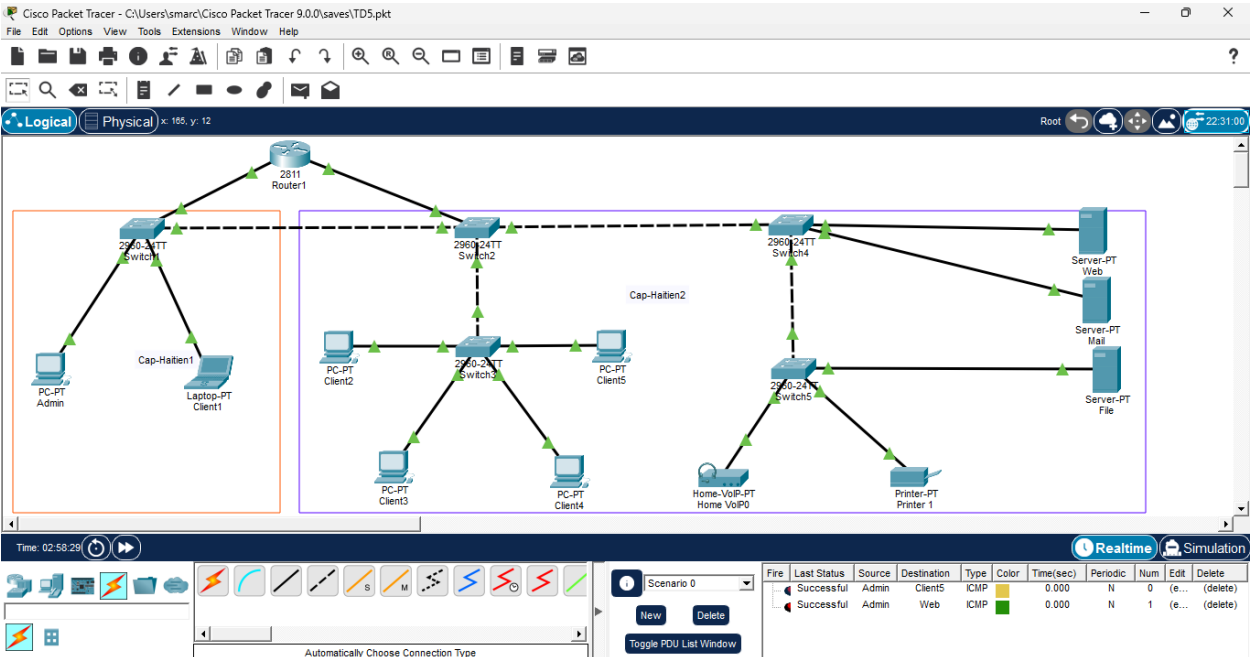
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=11ms TTL=127
Reply from 192.168.2.10: bytes=32 time=16ms TTL=127
Reply from 192.168.2.10: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 9ms

C:\>
```

Figure 1 Sur le PC Admin, j'ai épinglé 3 PCs il y a la connection des deux Vlan.

c) J'ai testé la connectivité des deux vlan en mode de simulation en temps réel



- Sur ces images, j'ai fait le test en mode simulation et le test est réussi avec succès.

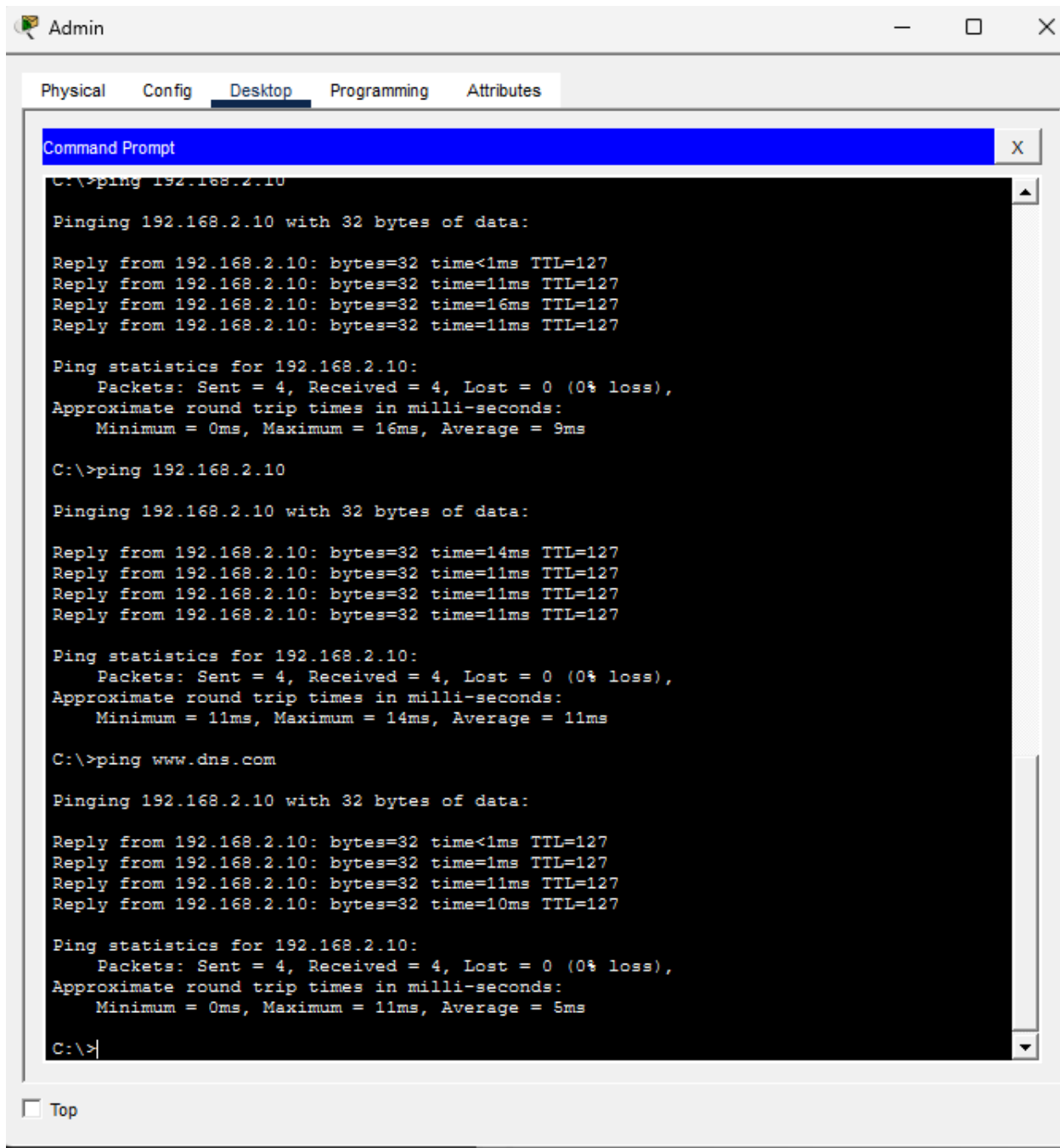
d) Configuration du serveur

The screenshot shows a web-based configuration interface for a server. The top navigation bar includes tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Services' tab is selected. On the left, a sidebar lists various services: SERVICES, HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP, and PRP. The 'DNS' service is selected. The main configuration area for DNS is displayed, showing the 'DNS Service' status as 'On'. Below this, the 'Resource Records' section is visible, with a 'Name' field and a 'Type' dropdown menu set to 'A Record'. An 'Address' field is also present. At the bottom of the configuration area, there are buttons for 'Add', 'Save', and 'Remove'. A table of resource records is shown, listing 9 records with columns for No., Name, Type, and Detail. The table is as follows:

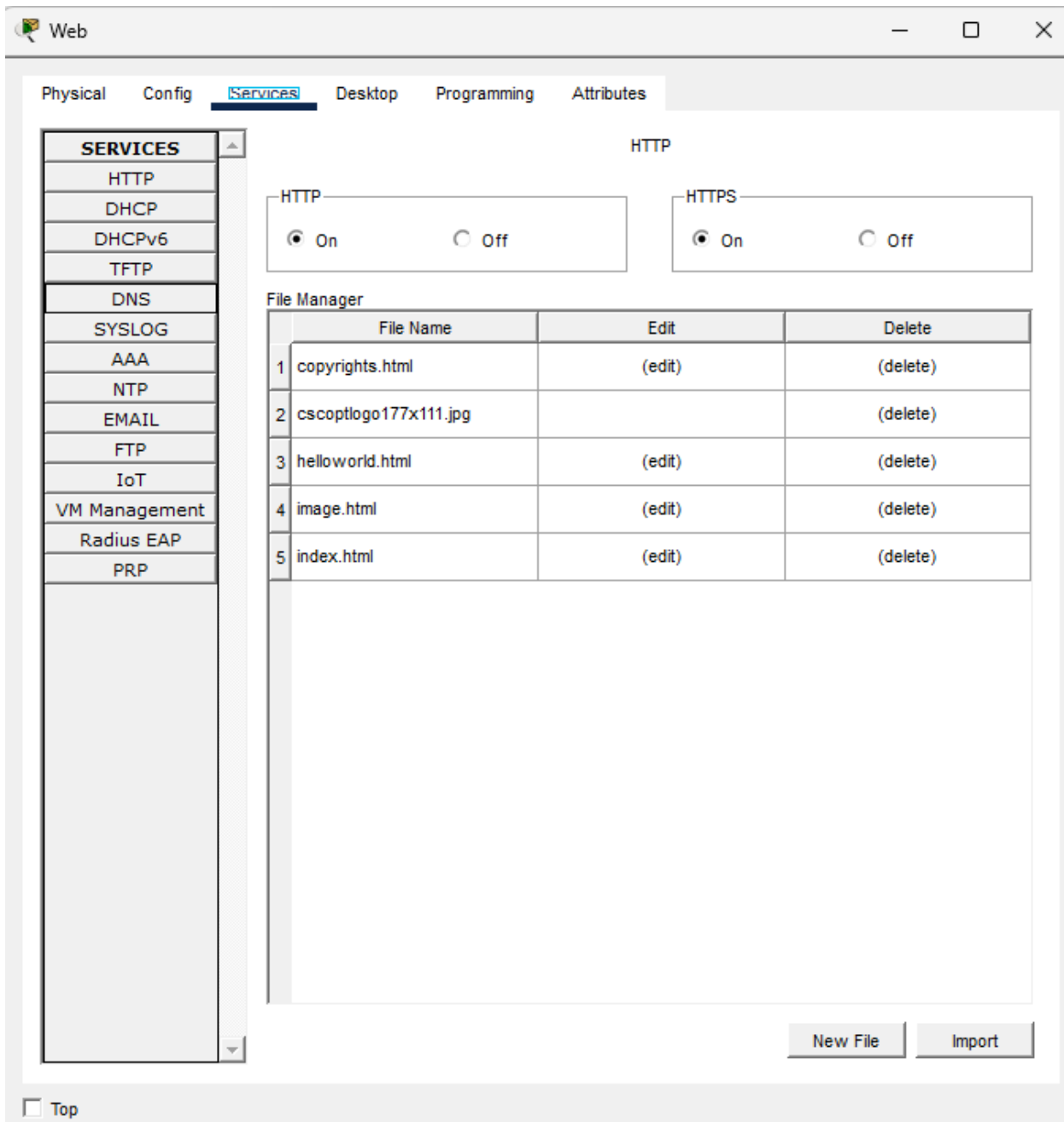
No.	Name	Type	Detail
0	admin	A Record	192.168.1.3
1	client1	A Record	192.168.1.4
2	client2	A Record	192.168.2.6
3	client3	A Record	192.168.2.7
4	client4	A Record	192.168.2.8
5	client5	A Record	192.168.2.9
6	server mail	A Record	192.168.2.11
7	www.dns.com	A Record	192.168.2.10
8	www.ius.com	A Record	192.168.2.12

At the bottom of the configuration area, there is a 'DNS Cache' button. The bottom of the interface features a 'Top' button.

- Sur ces images, j’ai rempli les champs nom et adresse de chaque topologie.



- Sur cette image j'ai effectué un ping sur le pc Admin afin de vérifier la connectivité.



- Sur ces images, j'ai pu cliquer sur File Manager pour modifier (edit) ou créer un nouveau fichier afin de passer à la page suivante..

Web

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

File Name: index.html

<html>

<center>Cisco Packet Tracer</center>

<hr>Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

<p>Quick Links:

A small page

Copyrights

Image page

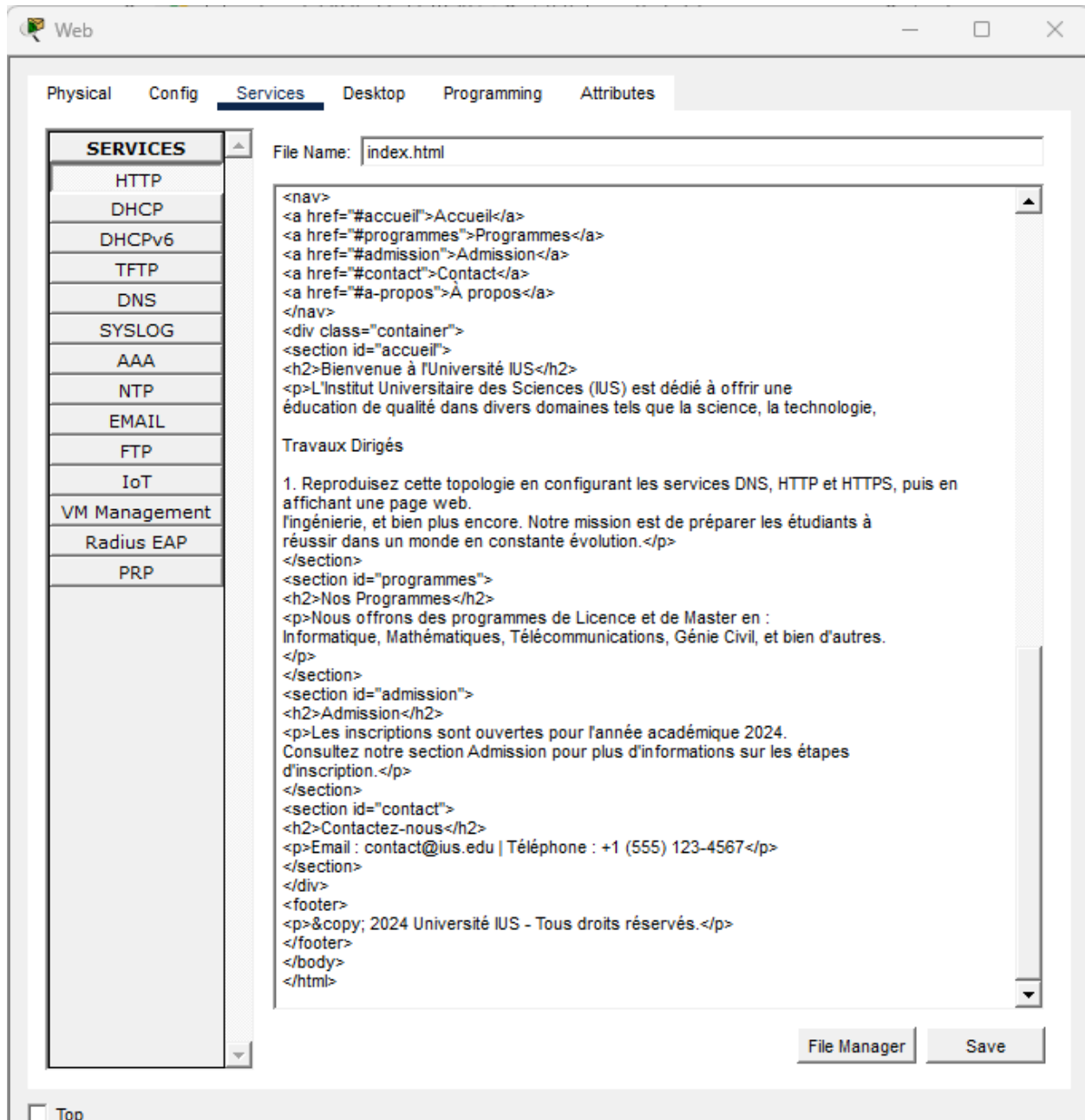
Image

</html>

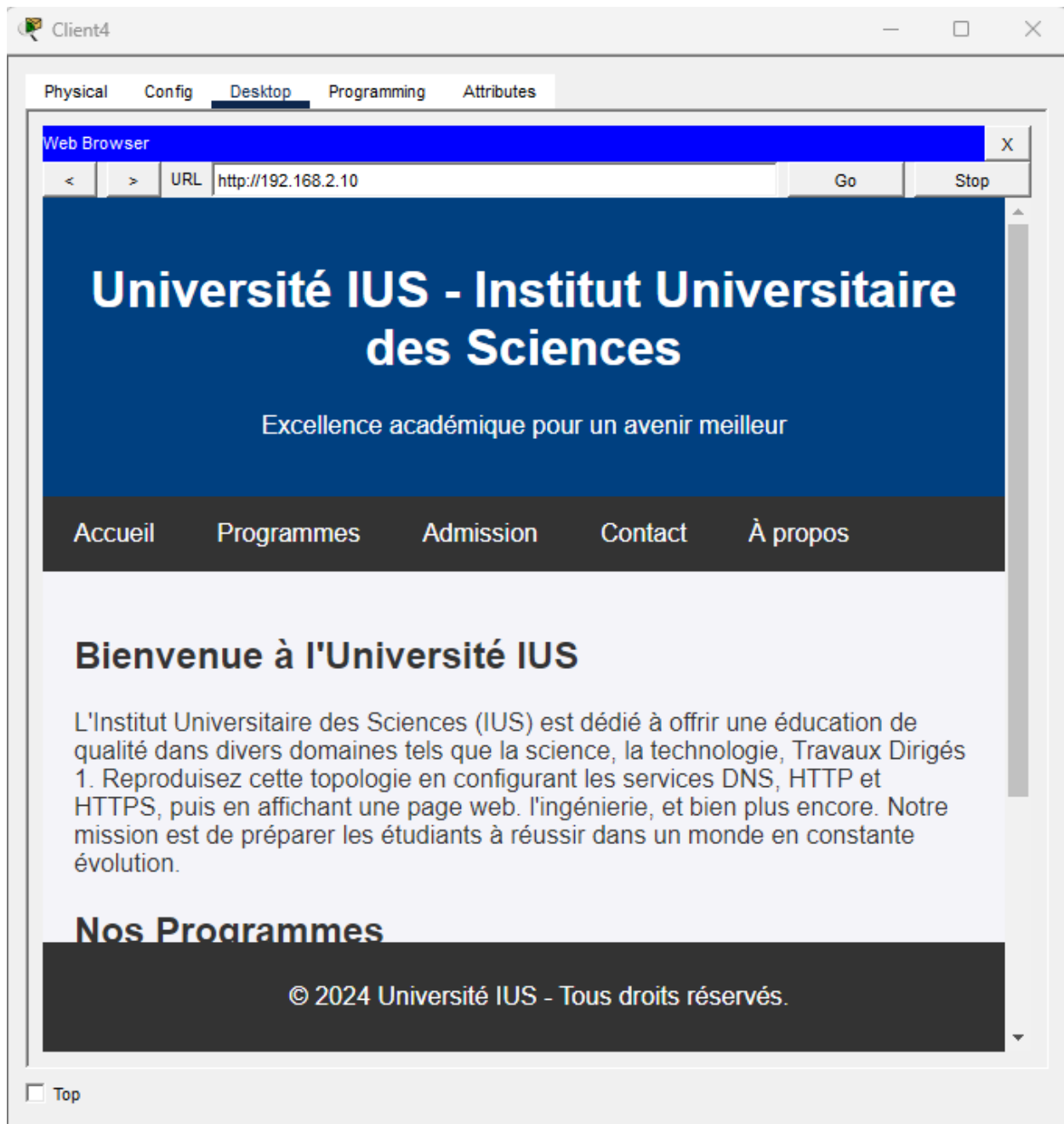
File Manager

Save

☐ Top



- Sur ces images, j'ai effectué un copier-coller afin d'ajouter le code (voir pages 44 et 45).



- Sur cette image, j'ai affiché le server Web.

Conclusion,

En dépit de tout, ce travail dirigé nous a permis de mettre en pratique les notions vues en cours concernant la configuration d'un réseau. Nous avons réussi à reproduire la topologie demandée, configurer correctement les services DNS, HTTP et HTTPS, et afficher une page web depuis les postes clients.

Cette méthode m'a aidé à mieux comprendre le fonctionnement des services réseaux ainsi que l'importance de la configuration correcte des équipements pour assurer une bonne communication dans un réseau.